University of Illinois Urbana Champaign

Convex Optimization

IE 521

Final Report: An Algorithm for Quadratic Programming

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Abstract

Frank and Wolfe's 1956 paper "An Algorithm for Quadratic Programming" introduced what is now known as the Frank–Wolfe or conditional gradient method, originally for concave quadratic programs with linear inequality constraints.[1]

1 Introduction and Research Context

The problem studied by Frank and Wolfe is the maximization of a concave quadratic function over a polyhedral feasible region. Their primal problem (PI) can be written as

$$\begin{aligned} \max_{x \in \mathbb{R}^n} \quad & f(x) := p^\top x - x^\top C x \\ \text{s.t.} \quad & x \geq 0, \quad Ax \leq b, \end{aligned} \tag{1}$$

where $C \succeq 0$ is symmetric positive semidefinite, so that $x \mapsto x^\top C x$ is convex and f is concave.[1]

By the mid 1950s, linear programming was already well developed: Dantzig's simplex method gave an algorithm for large LPs, and it was shown LP solutions can be found among the extreme points of a polyhedron. Nonlinear programming theory, including generalized Lagrange multipliers and the Karush–Kuhn–Tucker (KKT) conditions, was also available, but efficient algorithms for quadratic programming (QP) were not developed yet. Frank and Wolfe's goal is to design a method "well adapted to high-speed machine computation" for QP, reusing simplex-type technology.[1]

From a modern convex-optimization viewpoint, (1) is a convex optimization problem in the sense of maximizing a concave objective over a convex set. Frank and Wolfe's algorithm is one of the earliest ones for such problems and is a projection-free first-order method.[1]

2 Identifying Contributions

2.1 What is new in the work?

(1) Lagrange-multiplier reformulation via $g(x, u)$. Frank and Wolfe introduce the concave quadratic

$$g(x, u) = \nabla f(x)^\top x - ub = p^\top x - ub - x^\top C x, \tag{2}$$

where $u \geq 0$ are multipliers for the linear constraints $Ax \leq b$. They show that g can get us the solutions for (PI) through three criteria: a solution criterion, a boundedness criterion, and an approximation criterion that relates $g(x, u)$ to the optimal value of PI.[1]

(2) Auxiliary quadratic problem (PII). Using g and generalized Lagrange multipliers, they construct an auxiliary quadratic program (PII) in the variables (x, u) (and slack variables) whose solutions simultaneously yield:

- an optimal solution $x^\star$ of PI,
- a dual vector $u^\star$ certifying optimality,
- a feasibility and boundedness test for the original problem.

The QP is transformed into to another QP whose construction allows us to use LP algorithms as a subroutine.[1]

(3) Gradient-and-interpolation algorithm. They design an iterative method that:

1. uses the simplex method as a subroutine to solve linear subproblems over the feasible region,
2. computes convex combinations of current iterates and basic feasible points of PII,
3. terminates in a finite number of iterations for the quadratic case.

[1]

(4) Extension to general concave programming. In Section 6 of their paper, Frank and Wolfe extend the method to maximize a general concave, twice continuously differentiable function f over the same polyhedron. Under a boundedness assumption on the linearization $\nabla f(x)^\top w$ (for feasible w), they find gloval convergence for the algorithm and an $O(1/k)$ error. Finite termination is lost though.[1]

2.2 Research context and focus

The work is primarily theoretical. The paper does not focus on large numerical experiments or domain-specific applications.

3 Impact

3.1 What can we do now that we could not before?

The algorithm makes several things possible:

- **Algorithmic solution of QP using LP methods.** The method reduces concave QP to a sequence of LP subproblems plus line searches. Allowing one to modify existing LP algorithms for their own needs to solve QPs.[1]
- **Feasibility and boundedness certification.** Through the boundedness criterion, the feasibility of the constraints in the (x, u, slacks) system is the same as the boundedness of the original QP. If no such (x, u) can be found, the objective is unbounded above.[1]
- **Error bounds.** The approximation criterion implies that for any feasible (x, u),

$$M - f(x) \leq -g(x, u), \tag{3}$$

  where M is the optimal value. Meaning $-g(x, u)$ is an upper bound on the suboptimality.[1]

3.2 Theoretical implications

From a theoretical perspective, the paper:

- Shows how optimality conditions can be encoded in a quadratic function $g(x, u)$ and linear constraints on (x, u), slacks, and multipliers;
- Show the alogithm has $O(1/k)$ convergence rate for the objective gap.[1]

3.3 Applied and societal impact

While the original paper itself is not mentioning specific applications, QP has applications in finance, engineering, and machine learning.[2] Such as:

- **Portfolio optimization:** Markowitz-type mean–variance models are quadratic programs with linear constraints.
- **Engineering design and resource allocation:** There are design and allocation problems involving quadratic costs with linear feasibility constraints.
- **Modern ML tasks:** structured SVMs, low-rank matrix completion, and traffic equilibrium problems can be formulated or approximated as convex programs where this algorithm can be used to avoid projection methods.

4 Conclusion

Frank and Wolfe's paper is an old work in convex optimization. It introduced a computationaly nice algorithm for quadratic programming that reuses LP methods. The author just left it in the footnotes but there was an assumption the the LP subroutines are non-degenerate, though if the LP subroutine is degenerate then one must act accordingly (pivoting rules or other methods).

References

M. Frank and P. Wolfe. An algorithm for quadratic programming. *Naval Research Logistics Quarterly*, 3(1–2):95–110, 1956.

S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.